MIL-S-46128B(MR)
5 February 1982
SUPERSEDING
MIL-S-46128A(MR)
30 July 1975

MILITARY SPECIFICATION

STEEL, ALLOY, BARS, (V-MODIFIED 4330) FOR CRITICAL APPLICATION

This specification is approved for use by the US Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers wrought alloy steel bars produced to high quality standards.

1.2 Classification.

1.2.1 Form and condition. Unless otherwise specified in the contract or order (see 6.2), material shall be furnished as round bars in the cold-finished condition.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. Unless otherwise specified (see 6.2), the following standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-48 - Tolerances for Steel and Iron Wrought Products
FED-STD-66 - Steel; Chemical Composition and Hardenability
FED-STD-151 - Metals; Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-SSS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement there to, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E112 - Estimating the Average Grain Size of Metals

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC.

AMS 2300 - Premium Aircraft Quality Steel Cleanliness, Magnetic Particle Inspection Procedure
AMS 2640 - Magnetic Particle Inspection

(Copies of Aeronautical Material Specifications may be obtained from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, Pennsylvania 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material quality. When inspected magnetically in accordance with the procedures of AMS 2300, the maximum average frequency rating (F) shall not exceed 0.10 inch and the maximum average severity rating (S) shall not exceed 0.20 inch (see 4.5.1).

3.2 Manufacturing process. Unless a particular process is specified in the procurement documents (see 6.2), steel shall be manufactured only by the vacuum arc consumable electrode process in the remelt cycle except that steel cast as rectangular ingots for slab and plate may be produced by electroslag remelting.

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3.3 Chemical composition. The chemical composition shall normally be as specified in table I. When steel with a composition different than that shown in table I is submitted by the supplier, the steel product shall be subject to such tests as the purchaser requires for acceptance of the product.

TABLE I. Chemical composition

Element	Percent
Carbon	0.28 - 0.33
Manganese	0.65 - 1.00
Silicon	0.15 - 0.35
Nickel	1.65 - 2.00
Chromium	0.75 - 1.00
Molybdenum	0.35 - 0.50
Phosphorous	0.015 max.
Sulphur	0.015 max.
Vanadium	0.05 - 0.10
Copper	0.35 max.

3.3.1 Check analysis tolerances. Product analysis tolerances shall be in accordance with FED-STD-66.

3.4 Grain size. The austenitic grain size shall be predominantly No. 5, or finer, with occasional grains as large as No. 3 permissable.

3.5 Identification. Each bar shall be printed, stamped, or otherwise legibly marked in constantly recurring symbols at intervals not greater than 3 feet throughout the length or the bar. Markings shall include the heat number of the metal. Other markings shall be at the option of the manufacturer or as specified in the contract or order (see 6.2).

3.6 Dimensions. Dimensions shall be as shown in the contract or order or on the applicable drawings (see 6.2).

3.7 Tolerances. Unless otherwise specified (see 6.2), the dimensional tolerances shall conform to the limits of FED-STD-48, applicable to cold-finished steel.

3.8 Workmanship. Material shall be sound, of uniform quality and condition, and free from pipes, and shall be free from laps, cracks, twists, seams, or other defects detrimental to the fabrication and serviceability of the material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Acceptance tests. All of the required tests are classified as acceptance tests, for which necessary sampling techniques and methods of testing are specified in this section.

4.3 Examination of product. Sufficient spot checks shall be made to ensure compliance with the requirements for surface condition, identification (see 3.5), dimensions (see 3.6), and workmanship (see 3.8).

4.4 Sampling.

4.4.1 For magnetic particle inspection. Unless otherwise specified (see 6.2), samples for magnetic particle inspection shall be taken in accordance with AMS 2300.

4.4.2 For chemical analysis. One sample for product analysis shall be selected to represent each heat of steel in accordance with the requirements of Fed. Test Method Std. No. 151, for bars in small section. Samples shall weigh at least 57 grams (2 ounces).

4.4.3 For grain size. One or more samples shall be selected to represent each heat of steel from which material is submitted for acceptance.

4.5 Tests.

4.5.1 Magnetic particle inspection. Inspection and rating of specimens shall be performed in accordance with the procedures of AMS 2300 and AMS 2640.

4.5.2 Chemical analysis. Chemical analysis shall be conducted in accordance with Method 111 or 112 of Fed. Test Method Std. No. 151. In case of dispute, analysis by Method 111 shall be the basis for acceptance or rejection.

4.5.3 Grain size. The austenitic grain size shall be determined in accordance with ASTM E 112.

4.6 Rejection and retest. Unless otherwise specified in the contract or order (see 6.2), rejection and retest shall be conducted in accordance with the general section of Fed. Test Method Std. No. 151.

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5. PACKAGING

5.1 Preservation. Cleaning, drying and preservation shall be in accordance with manufacturer's commerical practice.

5.2 Packing.

5.2.1 Level C (Commercial). Packing shall be in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), shipments shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. Steel covered by this specification is intended for applications where high-shear-strength and high-notch-toughness are required.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Surface condition, if not cold finished (see 1.2.1).
(c) Manufacturing process, if other than as specified herein (see 3.2).
(d) Additional identification marking, if required (see 3.5).
(e) Exact lengths and length tolerances, if mill lengths are not acceptable (see 3.6).
(f) Diameter or cross-sectional dimensions of bars (see 3.6).
(g) Dimensional tolerances if other than cold finish tolerances of Fed. Std. No. 48 are applicable (see 3.7).
(h) Number of pieces for magnetic particle inspection, if other than as specified herein (see 4.4.1).
(i) Rejection and retest procedures when not in accordance with Fed. Test Method Std. No. 151 (see 4.6).
(j) When special marking for shipment requirements are necessary (see 5.3).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER | 2 DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION

b ADDRESS *(Street, City, State, ZIP Code)*

4 TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional

b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

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